

FOUR-MINUTE FLOW

0:00	POSITION	Local-first inspection assistant; selective VLM; human final authority.
0:20	PROBLEM	Consistency, scarce industrial data, and cloud cost/privacy friction.
0:50	ARCHITECTURE	Capture > OpenCV > YOLO classifier > decision > human. Uncertain branch only.
1:20	LIVE DEMO	Run NORMAL. Point: quality, class/confidence, ACCEPT, VLM not invoked, stored record.
2:15	VISUALIZER	Show RUST trace. Evidence view, not live 3D or dimensional measurement.
2:40	EVIDENCE	31/33 harder views; two rust views predicted normal. State limitation.
3:10	ECONOMICS	Cost-controlled architecture; pilot must measure ROI.
3:30	PILOT	Independent parts, fixed capture, risk metrics, one assisted cell.
3:48	ASK	Factory access, samples, defect expertise, pilot mentorship.

NUMBERS TO REMEMBER

31 / 33	Harder excluded photographs correct - 93.94%, not unseen-bearing accuracy.
7 / 7	Normal harder views correct.
7 / 7	Displaced harder views correct.
17 / 19	Rust harder views correct; two were predicted normal.
41-46 ms	Warm requests in one rehearsal; first request includes model initialization.
3 classes	Normal, displaced, rust. Missing-ball is excluded.

PRIMARY DEMO FILES

NORMAL	d1c83...	~96% > ACCEPT > no VLM
DISPLACED	974fdf...	~97% > REJECT > no VLM
RUST	7f8e3e...	~77% > REJECT > no VLM
UNCERTAIN	6df222...	~53% > REVIEW > fallback/VLM

DEFEND THESE BOUNDARIES

- Current YOLO checkpoint classifies the whole frame; it does not localize each defect.
- POC evidence is correlated with a few physical bearings; no production-accuracy claim.
- No missing-ball claim. Future method: detect each visible ball, then count and check angular gaps.
- No millimetre claim without fixed optics, calibration, reference dimension, and stable fixture.
- VLM advice is reversible and advisory. It never certifies a bearing or overrides the inspector.
- Cost is controlled by local inference and selective calls; ROI is a pilot measurement, not a current rupee claim.

15-SECOND RECOVERY

SLOW	First inference loads the model; warm requests were 41-46 ms in rehearsal.
CAMERA	Use the same local real photograph to isolate the pipeline from the driver.
GEMINI	Local path continues; uncertain case uses labelled fallback plus human review.
FAIL	Open backup video immediately. Do not debug on stage.
SURPRISE	Variation is why confidence is exposed and human authority remains.